

EUROPEAN CURRICULUM VITAE

PERSONAL INFORMATION

Name **Agostino Manzato**



Personal address **I-34170 Gorizia (GO)
Italy**

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I-33057 Palmanova (UD)
Italy**

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Nationality **Italian**

Civil Status **married**

Date of birth **16.08.1968**

WORK EXPERIENCE

- Dates (from – to) **01.07.2000 - now**
- Name of organization **Agenzia Regionale per la Protezione dell'Ambiente - Osservatorio Meteorologico Regionale
Regional Agency for Environmental Protection - Regional Meteorological Observatory**
- Type of organization **Public agency**
- Organization addresses **OSMER – ARPA FVG
via Natisone, 43
I-33057 Palmanova (UD)
Italy
Tel. +39 0432 926843**
- Type of business or sector **Meteorology**
- Position held **Meteorologist in permanent position**

- Main activities
 - radio-sounding analysis and atmosphere instability evaluation;
 - thunderstorm, rain and hail forecasting using Neural Networks;
 - linear and non-linear downscaling and statistical adaptation;
 - forecast verification for binary and continuous variables;
 - visualization of meteorological data for real time and case studies;
 - satellite analysis and instability evaluation of IASI vertical retrievals.
- Dates (from – to) **20.11.1996 - 30.06.2000**
- Name of organization Centro Servizi Agrometeorologici
Center for Agrometeorological Services
- Organization addresses CSA
Via Carso, 3
I-33052 Cervignano del Friuli (UD)
Italy
- Type of organization Public agency
- Type of business or sector Meteorology, weather forecasts
 - Position held Meteorologist in permanent position
 - Note I started working in the “*Thunderstorm and Hail Prevention Research Center*”, under the scientific direction of dr. Griffith Morgan. “CSA” became “OSMER – ARPA FVG” on 01.07.2000

EDUCATION AND TRAINING

- Dates **23.02.1994**
- Name of organization Università degli Studi di Padova
Padua University - Italy
- Subject Laurea in fisica
Degree in physics
- Title of qualification awarded Dottore in fisica (104/110)
Italian university degree in Physics (104/110)
- Further information Thesis titled: *Numerical simulations of the origin of the structures in the large scale universe*, prof. Sabino Matarrese relator and dr. Lauro Moscardini co-relator.

SCHOOLS AND WORKSHOPS

- Dates **October 1997**
- Name of organization HMZ - EUMETSAT - Ljubljana - Slovenia
- Subject Satellite meteorology and numerical weather prediction workshop
- Dates **April - May 1998**
- Name of organization NSSL, SPC and Oklahoma State University in Norman OK and NCAR in Boulder CO - USA
- Subject I made a one month long "training visit" to these important US Meteorological Centers where I met many researchers (H. Brooks, C. Doswell, D. Zrnica; J. Dye, C. Knight, J. Corbett,...)
- Dates **June 1998**
- Name of organization International Centre for Theoretical Physics - Trieste - Italy
- Subject Summer colloquium on the physics of the weather and climate “The effect of the topography on the atmospheric circulation” (featuring also prof. F. Mesinger)
- Dates **June 1999**

Name of organization	NCAR, Boulder CO - USA
Subject	Summer Colloquium on Ice formation in the Atmosphere (featuring also prof. W. Al Cooper)
Dates	June - July 2002
Name of organization	ISAC, CIMA, University of Genoa and Basilicata University, Saint-Oyen (Aosta) - Italy
Subject	Grand Combin Summer School on Physics and Predictability of Rainfall and Floods (featuring also prof. K. Emanuel)
Dates	July 2004
Name of organization	University of Trento, Trento - Italy
Subject	Summer School on Mountain Meteorology "Orographic Effects on Precipitation" (featuring also dr. R. Rotunno)
Dates	July 2005
Name of organization	CISM, OSMER, Udine - Italy
Subject	CISM Advanced School on Atmospheric Convection: Research and Operational Forecasting Aspects (featuring also prof. P. Markowsky)
Dates	January 2006
Name of organization	American Meteorological Society, Atlanta (GE) - USA
Subject	Short Course on Artificial Intelligence Applications to Environmental Science (featuring also prof. C. Marzban)
Dates	January 2006
Name of organization	American Meteorological Society, Atlanta (GE) - USA
Subject	86th AMS Annual Meeting
Dates	September 2007
Name of organization	EUMETSAT, ICTP, Trieste - Italy
Subject	EUMETSAT Training Workshop on Interpretation of Satellite Images/Products of Convective Storms (featuring also prof. D. Rosenfeld)
Dates	May 2008
Name of organization	SAR, ARPA Sardegna, ISAC, Alghero - Italy
Subject	Third Mediterranean School on Mesoscale Meteorology: "Fundamentals and applications of probabilistic forecasts and Ensemble Prediction Systems" (featuring also dr. T. Hamill)
Dates	March 2011
Name of organization	EUMETSAT, Darmstadt - Germany
Subject	Evaluating atmospheric instability from high spectral resolution IR satellite observations (featuring dr. P. Antonelli)
Dates	September 2012
Name of organization	CNR-ISAC, Castro Marina – Italy
Subject	Fourth ISAC Summer School: "Severe Convection Weather: Theory and Applications" (featuring also prof. E. Fedorovich)
Dates	August 2013
Name of organization	CISM – ISAC, Udine - Italy
Subject	The Fluid Dynamics of Climate (featuring also prof. M. Ghil)
Dates	March 2014
Name of organization	OSMER – ISAC, Visco - Italy
Subject	Previsioni di Limited Area Model in casi di Severe Weather (featuring also dr M. M. Miglietta)
Dates	June 2016
Name of organization	CNR-ISAC, Castro Marina – Italy
Subject	Sixth ISAC Summer School: "Advances in severe weather analysis: models and observations" (featuring also prof. H. Bluestein)

Dates	April 2017
Name of organization	OCCC – Bern University, Bern - Swiss
Subject	Second European Hail Workshop (scientific committee member)
Dates	October 2017
Name of organization	NERSC, CINFAI, ISAC – CNR headquarters, Bologna - Italy
Subject	Numerical Modeling, Predictability and Data Assimilation in Weather, Ocean and Climate

PERSONAL SKILLS AND COMPETENCES

MOTHER TONGUE **ITALIAN**

OTHER LANGUAGES

ENGLISH

- Reading skills Very good
- Writing skills Very good
- Verbal skills Good

SLOVENIAN

- Reading skills Good
- Writing skills Sufficient
- Verbal skills Sufficient

ORGANIZATIONAL SKILLS, REVIEW AND MEMBERSHIP On 18 September 2002 I have organized Seminar on Orographic Precipitations with dr. R. Rotunno and prof. R. Steinacker at OSMER; on 18 March 2014 I have organized a Seminar on Severe Weather Local Area Model forecasts with dr. Mario Marcello Miglietta and Silvio Davolio at OSMER. I have made reviews of scientific papers for many journals, including also *Journal of Geophysical Research*, *Atmospheric Research*, *Journal of Applied Meteorology and Climatology*, *Journal of the Atmospheric Sciences*, *Quarterly Journal of the Royal Meteorological Society*, *Atmospheric Science Letters*, *Meteorological Applications*, *Theoretical and Applied Climatology*, *Meteorology and Atmospheric Physics*, *International Journal of Climatology*, *Natural Hazards and Earth System Sciences*, *PLOS ONE* and *Atmosphere MDPI*.

I'm member of the *Convective Working Group* and of the *EUMETNET-NOWCASTING Activity group*.

TEACHING EXPERIENCE In 2003-2005 I was lecturer for the Applied Meteorology course at the University of Udine - Italy in the frame of the Computational Physics laurea (Master degree).

In 2006 I was tutor of a student during his thesis work in the frame of the master degree in Mathematics at the University of Udine - Italy.

On 17/11/2015 I was invited by ARPA Veneto to hold a seminar on atmosphere instability and sounding-derived instability indices in Teolo (PD).

On 04/10/2016 I was invited by ARPA Liguria to hold a seminar on nowcasting techniques in Genova.

In 2017 I was tutor of a student during his thesis work on hailpads analysis in the frame of the bachelor degree in Environmental Sciences at the University of Udine - Italy.

In 2018 and 2019 I was tutor of two students of the Udine University (Department of Mathematics, Computer Science and Physics) during their thesis on thunderstorm forecasting with convolutional neural networks (CNN) and analogue methods (AtmoSwing).

On 7/11/2017 I was invited by Istituto Cavanis of Venice to hold an advance meteorology course on the thermodynamic profile of the atmosphere.

On 20/03/2018 I was invited to hold a seminar on atmosphere instability at CETEMPS – University of L'Aquila.

TECHNICAL SKILLS
AND COMPETENCES

Computer knowledge at good level (like system administrator):
Linux (RedHat, SuSe, Ubuntu), Unix (SGI Irix), Windows (98, XP, Vista, 7, 10).
Programming languages at good level:
Python, Pytorch, Shell Script (BASH, TCSH), R (statistical language). I used to program in Fortran and C many years ago.
Software tools and packages:
ZEBRA (NCAR visualization tool for GRIB and NetCDF data), SatSignal (for MeteoSat and MSG data), MSG_RGB, HTDIG (search engine), LaTeX, OpenOffice, GIMP, PhotoShop.
I have experience on stepwise regression and Artificial Intelligence techniques (e. g. Neural Networks and Convolutional NN) applied to meteorological forecast problems.
I manage the data format conversion in GRIB (using Lats library and CDO) and NetCDF.
I manage the OSMER internal Intranet. I'm a skilled photograph (analog and digital).

PUBLIC SELECTION AND CALL
MEMBER

In October and November 2003 I was commissioner in a public call to select some physicist employees for ARPA FVG.
In June 2009 I was commissioner in a public call to select 5 physicist employees for ARPA FVG.
In June 2009 I was commissioner in a public call to select one mathematic employee for ARPA FVG.
In December 2010 I was commissioner in a public call to select one physicist employee for ARPA FVG (INCA-CE project).
In December 2018 I was commissioner in a public call of FVG Civil Protection to select a private company for restoring the Fossalon di Grado dual Doppler radar.
In August 2020 I was commissioner in a public call to select three physicist employees for ARPA FVG.
In October 2021 I was commissioner in a public call to select one meteorologist employee for ARPA FVG.

SELECTED PUBLICATIONS

Corresponding author:

- Manzato, A., Serafin, S., Miglietta, M. M., Kirshbaum, D., and W. Schulz, 2022: A pan-Alpine climatology of lightning and convective initiation, *Monthly Weather Review*, 150, 2213-2230, <https://doi.org/10.1175/MWR-D-21-0149.1>
- Manzato, A., Riva, V., Tiesi, A. and Miglietta, M.M., 2020: *Observational analysis and simulations of a severe hailstorm in northeastern Italy*, *QJ R. Meteorol. Soc.* 146, 3587-3611.
- Manzato, A., Pucillo, A., and Cicogna, A., 2018: *Improving ECMWF-based 6-hours maximum rain using instability indices and neural networks*, *Atmospheric Research*, 217, 184-197.
- Manzato, A. and I. T. Jolliffe, 2017: *Behaviour of verification measures for deterministic binary forecasts with respect to random changes and thresholding*, *Quarterly Journal of RMS*, 143 (705), 1903-1915.
- Feudale, L. and A. Manzato, 2014: *Cloud-to-Ground Lightning Distribution and its Relationship with Orography and anthropogenic emissions in the Po Valley*, *Journal of Applied Meteorology and Climatology*, 53, 2651-2670.
- Manzato, A., Davolio, S., Miglietta, M. M., Pucillo, A., and M. Setvák, 2014: *12 September 2012: A supercell outbreak in NE Italy?* *Atmos. Res.*, 153, 98-118.
- Manzato, A., 2013: *Hail in NE Italy: A neural network ensemble forecast using sounding-derived indices*, *Weather and Forecasting*, 3-28.
- Manzato, A., 2012: *Hail in NE Italy: Climatology and bivariate analysis with the sounding-derived indices*, *Journal of Applied Meteorology and Climatology*, 51, 449-467.
- Manzato, A., 2008: *A Verification of Numerical Model Forecasts for Sounding-Derived Indices above Udine, NE Italy*, *Weather and Forecasting*, 23, n3 477-495.
- Manzato, A., 2007: *A Note on the Maximum Peirce Skill Score*, *Weather and Forecasting*, 22, n5, 1148-1154.
- Manzato, A., 2005: *The use of sounding derived indices for a neural network short-term thunderstorm forecast*, *Weather and Forecasting*, 20, n6, 896-917.
- Manzato, A. and G. M. Morgan, 2003: *Evaluating the sounding instability with the Lifted Parcel Theory*, *Atmos. Res.*, 67-68, 455-473.
- Not lead authored:
- Jl Yano, A Manzato 2022: *Does More Moisture in the Atmosphere Lead to More Intense Rains?*, *J. Atmospheric Sciences*, 79, 663-681.
- Miglietta, M., A. Manzato, R. Rotunno, 2016: *Characteristics and Predictability of a Supercell during HyMeX SOP1*, *Quarterly Journal of RMS*, 142, 2839-2853.
- Mecikalski, J., D. Rosenfeld, A. Manzato, 2016: *Evaluation of Geostationary Satellite Observations and the Development of a 1-2 hour Prediction Model for Future Storm Intensity*, *JGR*, 121, 6374-6392
- Davolio, S., A. Volontè, A. Manzato, A. Pucillo, A. Cicogna, M. E. Ferrario, 2015: *Mechanisms producing different precipitation patterns over North-Eastern Italy: insights from HyMeX-SOP1 and*

BIBLIOMETRIC INDICES

More than 30 scientific articles with more than 850 citations.

H-Index=19 (Google Scholar). Research Gate Research Interest Score= 427.

FURTHER DETAILS OF THE CURRENT ACTIVITY	Forecasting severe weather with sounding-derived indices and Artificial Neural Network or CNN: I have used the sounding-derived indices as input of ANN ensemble for short-range forecast of thunderstorm and rain occurrence and intensity. I have applied the same methodology to ECMWF model-derived pseudo-indices and also to the hail problem. I have evaluated the thunderstorm forecasting skill of a system based on IASI satellite-derived data. Recently, I'm testing Convolutional Neural Networks in a Pytorch environment for forecasting lightning activity. Downscaling: I have supervised a mathematic student during his thesis on downscaling of maximum and minimum daily temperatures from ECMWF data using a linear multiregression (MOS). Forecast Verification: I'm particularly interested in verification of the binary (or multi-categorical) classifiers using the Signal Detection Theory and in particular the maximum Pierce Skill Score. I'm also interested in combining classifiers in ensemble (bagging, boosting, MV,...). Climatology: In the last period, I studied the climatology of convective initiation from lightning data in the Alpine domain and the climatology of hailstones in NE Italy from hailpads..
FURTHER INTERESTS	Classical music, Hi-Fi, electronics, photography.
ADDITIONAL INFORMATION	I have a 15-year old daughter.
DATE	26 October 2022